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Applied Session: 7 | Teaching Associates: Michael Niemann & Ayden Zhou

Project Title: Programming Exercise 3: D3

Reflection:

Across each of the three programming exercises on Australian Parrot data, Tableau, R and D3 each offered different level of control over the final design choices of the visualization in my opinion which shaped my experience significantly.

Tableau quickly produced readable charts by mapping size and color, but lacked customization. I felt stuck with its default layouts, like fixed legends. Tableau made the decisions for me, making design feel more like a negotiation than a creative process.

R, which I used to build a Shiny app combining a grouped bar chart of observations by season and a Leaflet proportional symbol map. Implementing species checkboxes, season radio buttons, and hover tooltips showing observation count and state gave the app genuine interactivity. I also had variety of options in map styles and tiles which improved customization that Tableau lacked. There were few limitations; few design packages offered limited choices but compared to Tableau, it was miles better. What made R hard was that wrangling the data into the right shape before any of that was possible consumed most of my time.

D3 demanded the most upfront effort. Every element had to be constructed from scratch such as the log scale mapping observation counts from minimum 496 to maximum 5.3 million into readable node sizes, the per-species max-normalised bar glyphs, the ancestor-descendant highlight using `d.ancestors()` and `d.descendants()`, and the viewport-clamped tooltip. But that cost came with something the other tools could not offer: complete control over every visual decision. I was the artist without any restriction bound by a package or style. I could literally draw whatever I want wherever I want.

For a similar task in future I would choose D3. The output reflects exactly what I intended, not what the tool was willing to produce.

Originality Declaration

I used Claude to assist with writing the JavaScript code for all five task blocks in `template.html`. I reviewed, tested, and verified all generated code in the browser before submission on my own. AI-generated code was also manually checked for hardcoded values and corrected where data-derived values were more appropriate.

List of Prompts

- [Shared two screenshots] What about now? Also the Family, Bar glyph and Number of observations on the right hand side is a bit untidy don't you think? Its not uniform looking. "Neophema" looks small compared to other Genus.
- Uncaught Error: Uncaught ReferenceError: d3 is not defined — What's this?
- Check this against the spec let me know all criteria are met perfectly without leaving anything.
- This still has the same legends problem — its being congested.
- The line above Bar glyph and Number of observations is way too close to the actual words.
- Check if we have hardcoded anything which shouldn't be hardcoded.
- Is there any improvements needed for template.html?

The reflection was written entirely by me without the use of any Generative AI tool.